Day: Date:

Section - B (33 marks)

Q:2 (i)(or)

A:2(i)(or)

Non-physical quantities:

Non-physical quantities are those ~~ques~~ quantities which cannot be measured by ~~nur~~ numerical value (magnitude) or size. They are subjective and depend on personal perception or opinion.

Examples:-

:> ~~Tastare~~ Taste -

:> Feelings / emotions

:> Beauty / colour perception

Why they cannot be measured :-

:> Non-physical quantities lack a standard unit and numerical value. They are qualitive rather than quantitive - different people may have different perceptions of the same stimulus.

Q:2(ii)

A:2(ii)

Base quantity	symbol	SI - base unit	unit symbol
Length	L	meter	m
Mass	m	Kilogram	kg
Time	t	second	s

Day: Date:

Base quantity	symbol	S-I base unit	unit symbol
Electric current	I	ampere	A
Temperature	T	Kelvin	K
Amount of substance	n	mole	mol
luminous ~~Kity~~ intensity	I_v	candela	cd

Q:2(iii)

A.2(iii)

Scalar quantity	Vector quantity
	Defination
1) Physical quantity which can be measured by it's numerical magnitude (or size) with proper unit only are termed as scalar quantity	1) Physical quantity which not only requires numerical magnitude (or size) with proper unit but also direction are termed as vector quantities.
	Representation
2) Scalar quantities can be added, subtracted, divided and multiplied by using ordinary algebra.	2) Vector quantities cannot be added, subtracted, divided, and multiplied by using the usual rules of algebra. They follow their own set rule

	known as vector algebra.

Examples

Examples of scalar quantities are :-	Examples of vector quantities are :-
* Length	* Momentum
* ~~Time~~ Speed	* Velocity
* Mass	* Force
* Temperature	* Weight
* Energy	* Accelaration
* Distance	* Displacement.

Q:2 (iv)

A:2 (iv)

formula:- least count (L.C) = smallest main scale divisions ÷ N.o of vernier scale divisions

Data:- Smallest main scale division = 1
N.o of vernier scale divisions = 10

$$L.C = \frac{1}{10} = 0.1mm$$

$$\Rightarrow 0.1mm = 0.01cm$$

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Meanings:-

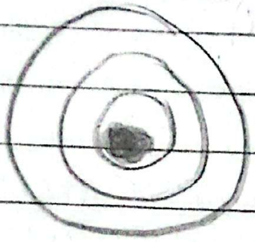
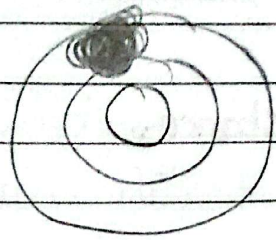
The least count represents the smallest measurement that the vernier caliper can make. A least count of 0.1mm, means the vernier caliper can measure ~~lenghts~~ lengths with a precision of 0.1 mm (one tenth of a millimeter) which is much more ~~a~~ precise than a simple metre rule.

Q:2(v)

A:2 (v)

Precision:-	Accuracy
Defination	
~~Phy~~ Precision can be ~~Lift~~ thought of as the degree of agreement between repeated measurements of the same quantity	Accuracy refers to how close a measured a value is to the true or accepted value.
Measurements	
Precision focuses on the ~~consistency~~ consistency and reproducibility of measurement.	An accurate measurements is one that is close to the true value, regardless of whether it is consistently reproducible.

Day: Example Date:

	
High precision and high accuracy	High precision and low accuracy.

Q.2(vi)

Q.2(vi)

a) 100.8 s.

Solution:-

Significant figures:- 4 significant figures.

Scientific notation:- $\underbrace{100}.8\,s$

$1.008 \times 10^{2}\,s$

b) 0.00580 km

Solution:-

Significant figures:- 3 significant figures.

Scientific notation:- $0.00\underbrace{5}80\,km$

$5.80 \times 10^{-3}\,km$

c) 210.0 g

Solution:-

Significant figures:- 4 significant figures.

Scientific notation:- $\underbrace{210}.0\,g$

$2.10 \times 10^{2}\,g$

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Day: Date:

Q: 2(vii)

A: 2 (vii)

Distance:-

The length of path travelled between two points called distance. Distance has no direction so it is a scalar quantity. It has a SI unit as meter (m).

Displacement:-

The shortest distance from initial position to final position (or straight directed distance) is called displacement. Displacement has direction so it is vector quantity. It has same SI unit (metre).

A runner ~~of~~ on a circular track:-

* Distance :- 400 m
* Displacement . 0,

reason:- The runners returns to the starting point. The initial and final positions are the same so the shortest distance (~~dis~~ displacement) ~~is zero~~ between them is 0 (zero).

Q:2(viii)(or)

Q:2(viii)(or)

Average Speed	Instatqneous Speed
Defination.	
* The average speed is the total distance covered in the passage of time.	The speed at any specific instout of time is called Instatoneous speed.
RePresentation	
* This calculation averages out any variations in speed throughout the journey.	If we are not Looking at speedometre of car, we only have rough idea of how fast we are moving and how much we should reduce speed.

Calculation:-

Distance : 70Km

Time :- 1hr

Speed = ? $= \frac{s}{t} = \frac{70}{1}$

$= 70 Km/h$

Converting km/hr to m/s

~~70 x 1000~~

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$= 70 \times \frac{1000}{3600}$

$= \frac{700}{36}$

$= 19.44\, m/s$

~~Q:2 (ix) (or)~~

A: 2 (ix) (or)

~~Data:~~
~~Distance:- 800~~
~~Time:-~~
~~Displacement:-~~

Data:-

Distance :- 30Km = $30 \times 1000 = 30{,}000\, m$

Time:- 7min 30s = $7(60) + 30 = 420 + 30$
$= 450s$

Speed in m/s :-

$v = \frac{s}{t} = \frac{30{,}000}{450}$

$= 66.67\, m/s$

Speed in Km/hr

Speed :- $30Km \div \frac{450}{3600}$

$= 30 \div 0.125h$

$= 240 Km/hr$

Q:2(x)(or)

A:2(x) (or)

Data:-

Initial velocity $(\vec{v})$ = 0 m/s
Final velocity $(\vec{v})$ = 26.8 m/s
Time = 3 seconds.

Formula:-

$$\text{acceleration} = \frac{\cancel{\text{initial}}\ \text{final velocity} - \cancel{\text{Final}}\ \text{initial velocity}}{\text{Total time}}$$

$$= \cancel{\frac{0 - 26.8\,m/s}{3\,s}}$$

$$= \cancel{-26.8\,m/s} \qquad \frac{26.8\,m/s - 0}{3\,s}$$

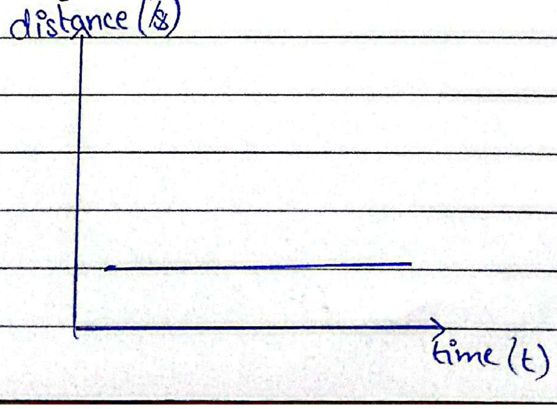
$$= \frac{26.8\,m/s}{3\,s}$$

$$= 8.93\,m/s^2$$

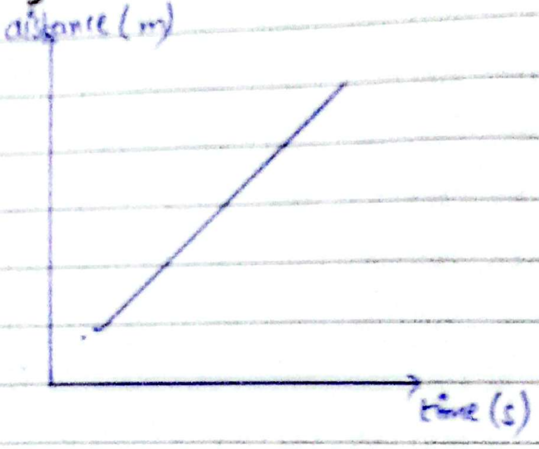
Q:2(xi)(or)

A:2(xi)(or)

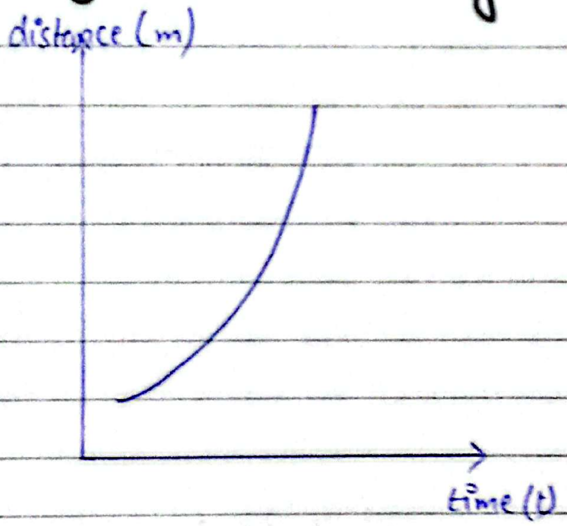
a) A body at rest :-



b) A body with constant speed:-



c) A body with increasing speed:-



Day: Date:

Section C

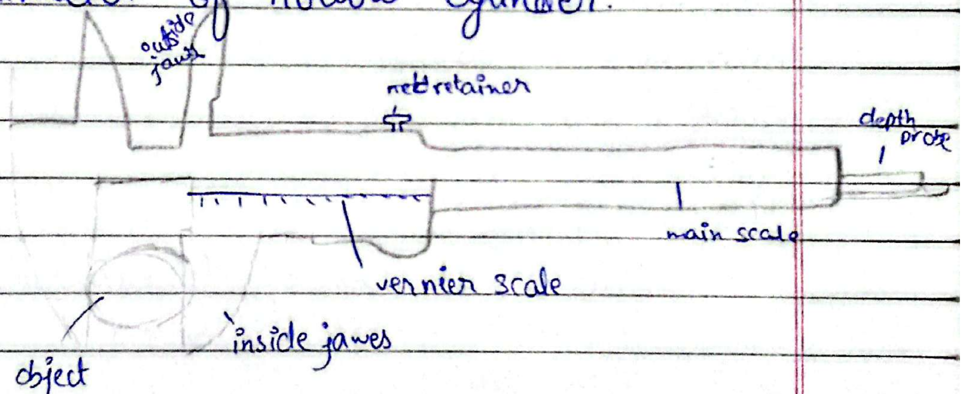
Q:3

A:

Vernier caliper:-

A vernier caliper is a tool for precise measurements, using a sliding secondary scale to measure fractions of the smallest division on the ~~mal~~ main scale.

It can be used to measure the thickness, diameter or width of an object and the internal, external diameter of hollow cylinder.



There are two scale on vernier caliper:-

Main scale :-

A main scale which has ~~Only~~ marking of usually of 1mm each and it contains jaw on it's left end.

Vernier scale:-

A Sliding scale called vernier scale which has markings of some multiple of the marking on the main scale.

L.C (Least count)

~~Maxim~~ Minimum length which can be measured accurately with the help of a vernier calipers called least count.

L.C can be obtained from dividing the value of smallest division on main scale by total number of divisions on vernier scale.

$$L.C = \frac{\text{Smallest division on main scale.}}{\text{Total number of divisions on vernier scale.}}$$

Zero error :-

Lack of zero error :-

There is no zero error as the zero line of vernier scale is conciding with the zero of vernier scale.

Positive zero error :-

Zero error is positive as vernier scale is towards the right of the zero of main scale.

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Negative zero error:-

Zero error is negative as vernier scale is towards the left of main scale.

Measurement of vernier calipers:-

* Note the least count for ~~vernier~~ vernier (It is usually written on vernier caliper or we can find by method). Determine and correct zero error ~~is~~ if any.

* Fix the small sphere in the jaws and note the complete divisions of main scale by the zero of vernier scale. This is main scale reading

* Look for the division of vernier scale that is conciding with any division on main ~~scale~~ scale. This is vernier scale reading.

* Multiply the vernier scale reading with least count which is the fraction to be added with main scale reading. This fraction will be smaller than the main scale least count, thus vernier caliper measure the reading to the part of millimetre.

Day: Date:

Q: ~~3~~ 4 (or)
A: 4 (or)

Scientific notation:-
Standard form ~~of~~ or scientific notation represents a number as the product of a number greater than 1 and less than 10 (called the mantissa) and a power of 10 (termed as exponent).
Mathematically,
number = mantissa x 10 exponent.

Why it is used in Physics:-

* Physics deals with extremely large and small numbers.

* Scientific notation make these numbers easier to read, write and understand.

* It reduces the chances of errors.

* It shows the number of significant figure.

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Conversion :-

a) 149,530,000,000 m
Scientific notation :-
1.4953×10^{11}.

Prefix :-
: $149.53 \times 10^{11} \times 10^{-2}$
: 149.53×10^{9} m
: 149.53 ~~giga~~ Giga m
: 149.53 Gm.

b) $\underbrace{0.0008}$ g

Scientific notation :-

8.0×10^{-4} g

Prefix :-

8.0×10^{-4} g
~~$80.0 \times 10^{8.4} \times 10^{11}$~~
$0.8 \times 10^{1} \times 10^{-4}$
0.8×10^{-3}
0.8 mg

b) 86,400 s
Scientific notation :-
: $\underbrace{86,400}$ s
: 8.64×10^{4}

- prefix :-
8.64×10^{4}
$86.4 \times 10^{4} \times 10^{-1}$
86.4×10^{3}

8,6.4 ~~x~~ Ks

Prefixes:-

Definition:-

Prefixes are words or symbols placed before SI units to represent decimal multiples or sub-multiples of those units.
They make very large or very small quantities easier to express.

Three large prefixes

Prefix	Symbol	Multiplier	Example
Giga	G	10^9 (1,000,000,000)	1 GHz = 10^9 Hz
Mega	M	10^6 (1,000,000)	1 MW = 10^6 W
Kilo	K	10^3 (1,000)	1 km = 10^3 m

Three small prefixes

Prefix	Symbol	Multiplier	Example
Milli	m	10^{-3} (0.001)	1mm = 10^{-3} m
Micro	μ	10^{-6} (0.000001)	1 μm = 10^{-6} m
Nano	n	10^{-9} (0.000000001)	1nm = 10^{-9} m

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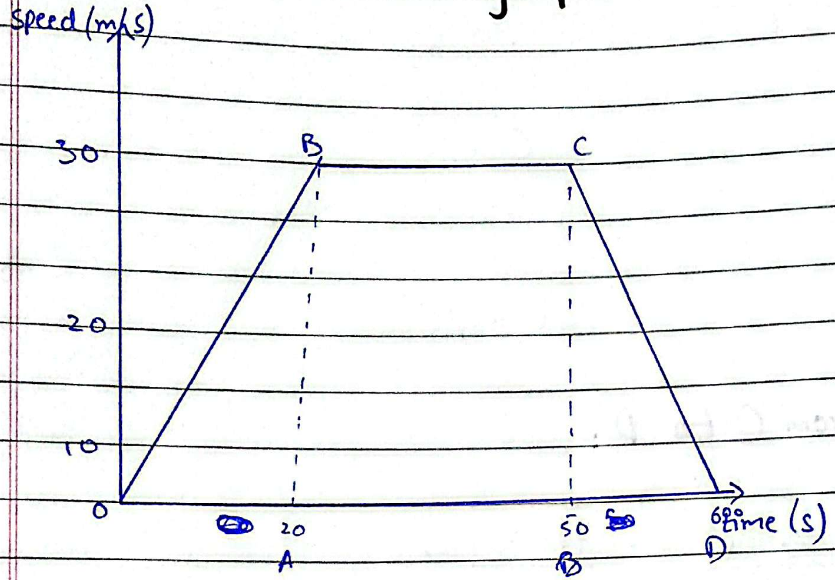
Q: 5 (or)
A: 5 (or)

a) ~~Distance~~ Speed - time graph



A to B = (0 - 20 s) = uniform accelaration
B to C = (20 - 50 s) = constant speed
C to D = (50 - 60s) = uniform accelaration.

b) Accelaration :-

A to B = $\frac{v_f - v_i}{t}$

$= \frac{30 - 0 \text{ m/s}}{20 s}$

$= \frac{30 \text{ m/s}}{20 s}$

$= 1.5 \text{ m/s}^2$

from B to C

$= \frac{v_f - v_i}{t} = \frac{30 - 30 \text{ m/s}}{30 \text{ s}}$

$= \frac{0 \text{ m/s}}{0 \text{ s}}$

$= 0 \text{ m/s}^2$

from C to D:-

$= \frac{v_f - v_i}{t} = \frac{0 - 30 \text{ m/s}}{10 \text{ s}}$

$= \frac{-3\cancel{0} \text{ m/s}}{1\cancel{0} \text{ s}}$

$= -3 \text{ m/s}^2$

c) Total Distance and Average speed:-

$\frac{1}{2}(bh) + L \times W + \frac{1}{2}(bh)$

$= \frac{1}{2}(20 \times 30) + (30 \times 30) + \frac{1}{2}(10 \times 30)$

$= \frac{1}{2}(600) + 900 + \frac{1}{2}(300)$

$$= \frac{\overset{300}{\cancel{600}}}{\cancel{2}_1} + 900 + \frac{300}{2}$$

$$= 300 + 900 + 150$$

$$= 1350$$

total tim. $20 + 10 + 30 = 60s$

Average speed:- $8v = \frac{s}{t} = \frac{135\cancel{0}}{6\cancel{0}}$

$$= \frac{135}{6}$$

$$= 22.5 m/s.$$

(Q:6)

A:6

Speed :-

Measure of the distance covered (Δs) ~~p~~ with passage of time is called speed. It is a scalar quantity with no direction.

Mathematically:-

$$v = \frac{\Delta s}{\Delta t} \quad \text{or} \quad v = \frac{s_f - s_i}{t_f - t_i}$$

Velocity:-

Measure of displacement ($\Delta\vec{s}$) with passage of time (Δt) is called velocity. It is a vector quantity with direction.

Mathematically,

$$\vec{v} = \frac{\Delta\vec{s}}{\Delta t} \quad \text{or} \quad \vec{v} = \frac{\vec{S}_f - \vec{S}_i}{t_f - t_i}$$

Accelaration:-

The measure of change in velocity ($\Delta\vec{v}$) with the passage of time (Δt) is called accelaration $\vec{a}$. It is a vector quantity same as velocity.

Mathematically,

$$\vec{a} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i} \quad \text{or} \quad \vec{a} = \frac{\Delta\vec{v}}{\Delta t}.$$

b) Shoib Aktar:-

Data:-

Speed = 161.3 Km/hr

Distance :- 12.5 m

Conversion

$$161.3 \times \frac{1000}{3600}$$

$$= \frac{161300}{3600} = 44.81\ m/s$$

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Time $= \frac{s}{v}$

$= \frac{17.5}{44.81}$

$= 0.39s.$

c) Block falling from Building

Data:-
Mass : 2 kg
Initial velocity: 0 m/s
Final velocity = 78.5 m/s
Accelaration: 9.8 m/s^2

Formula:-

$a = \frac{v_f - v_i}{t}$

rearranging formula:- $t = \frac{v_f - v_i}{a}$

$t = \frac{(78.5-0)}{9.8}$

$= \frac{78.5}{9.8}$

$= 8.01$

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Version: 1

Total Marks: 65 | **Time Allowed: 2 hours 30 minutes**

General Instructions:

1. Answer all questions from Section A (MCQs) on this page by filling in the correct bubble.
2. In Section B, attempt **any EIGHT (8)** short questions. Each question carries **3 marks**.
3. In Section C, attempt **any THREE (3)** long questions. Each question carries **5 marks**.
4. Use of scientific calculator is allowed.
5. Write neatly and legibly. Marks may be deducted for untidy work.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks) Time: 15 minutes

#	Question	A	B	C	D	Answer
1	Which one of the following is **not** a derived unit?	pascal	kilogram	newton	watt	Ⓐ ● Ⓒ Ⓓ
2	In SI, the base unit of temperature is:	celsius	kelvin	fahrenheit	joule	Ⓐ ● Ⓒ Ⓓ
3	The SI derived unit of force ($kg{\cdot}m/s^2$) is given the special name:	pascal	joule	newton	watt	Ⓐ Ⓑ ● Ⓓ
4	The mass of Earth (5,980,000,000,000,000,000,000,000 kg) in scientific notation is:	5.98×10^{24} kg	59.8×10^{23} kg	5.98×10^{25} kg	0.598×10^{25} kg	● Ⓑ Ⓒ Ⓓ
5	Which of the following prefixes represents the largest value?	mega	pico	peta	kilo	Ⓐ Ⓑ ● Ⓓ
6	Which of the following is a **vector** quantity?	distance	speed	displacement	mass	Ⓐ Ⓑ ● Ⓓ
7	The least count of a vernier caliper with smallest main scale division 1 mm and 10 vernier divisions is:	0.01 mm	0.1 mm	1 mm	0.5 mm	Ⓐ ● Ⓒ Ⓓ
8	The number of significant figures in 0.00580 km is:	2	3	5	6	Ⓐ ● Ⓒ Ⓓ
9	The motion of a rotating fan is an example of:	translatory motion	vibratory motion	random motion	rotatory motion	Ⓐ Ⓑ Ⓒ ●
10	A girl walks 3 km west and 4 km south. Her total distance and displacement are:	7 km, 7 km	1 km, 7 km	7 km, 1 km	7 km, 5 km	Ⓐ Ⓑ Ⓒ ●
11	In 5 s a car accelerates so that its velocity increases by 20 m/s. The acceleration is:	$0.25\ m/s^2$	$4\ m/s^2$	$25\ m/s^2$	$100\ m/s^2$	Ⓐ ● Ⓒ Ⓓ
12	The slope of a distance-time graph gives the:	velocity	acceleration	speed	displacement	Ⓐ Ⓑ ● Ⓓ